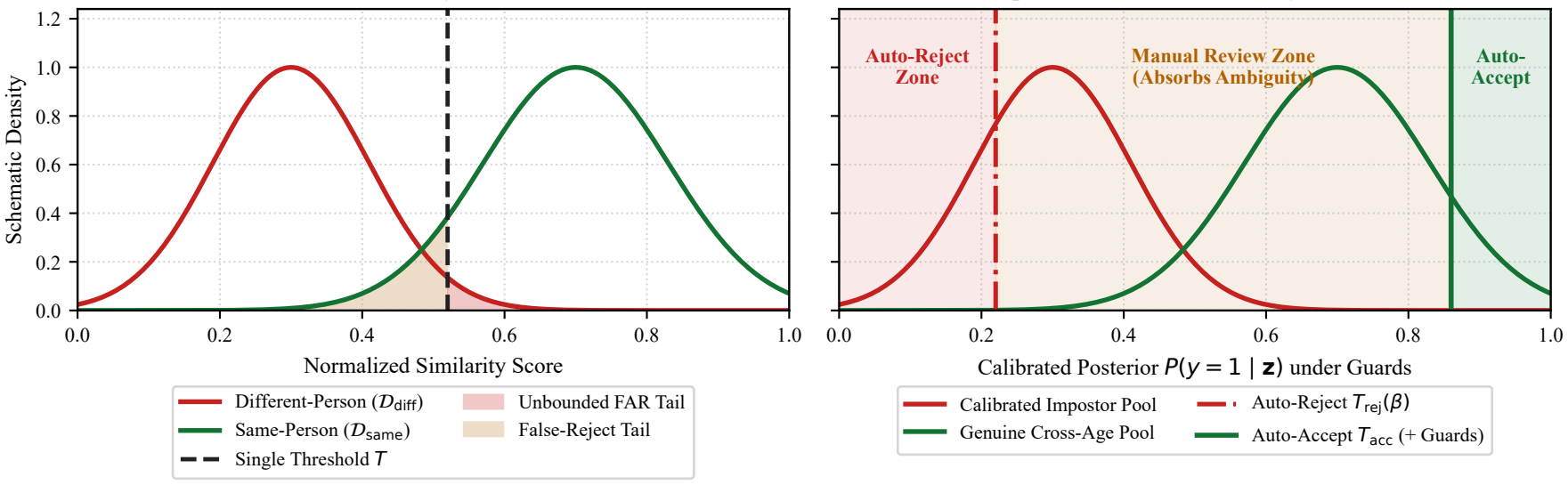

(a) Conventional Binary Threshold Matcher
Schematic Density
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Similarity Score
Different-Person (D_{diff})
Same-Person (D_{same})
Single Threshold T
Unbounded FAR Tail
False-Reject Tail
(b) Proposed Guarded Three-Way Decision Rule
Auto-Reject Zone
Manual Review Zone (Absorbs Ambiguity)
Auto-Accept
0.0
0.2
0.4
0.6
0.8
1.0
Calibrated Posterior $P(y = 1 \mid \mathbf{z})$ under Guards
Calibrated Impostor Pool
Genuine Cross-Age Pool
Auto-Reject $T_{\text{rej}}(\beta)$
Auto-Accept T_{acc} (+ Guards)